

CS 361: Countability and Diagonalization

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Definitions

A function $f : A \rightarrow B$ is called *one-to-one* (or *injective*) if different elements of A map to different elements of B . Formally, f is injective if and only if for all $x, y \in A$ such that $f(x) = f(y)$, then $x = y$. A function $f : A \rightarrow B$ is called *onto* (or *surjective*) if every element of B has at least one pre-image in A . Formally, f is surjective if and only if for all $b \in B$, there exists $a \in A$ such that $f(a) = b$. A function is *bijective* or a *bijection* if it is both one-one and onto.

A set S is finite if there is a bijection $f : S \rightarrow \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. We call n the *size* of S , denoted as $|S| = n$. S is *infinite* if no such bijection exists.

An set S is *countably infinite* if there exists a bijection $f : S \rightarrow \mathbb{N}$. We say a set S is *countable* if it is finite or infinite countably infinite. We say a set is *uncountable* if it is not countable. Example of countable sets include the natural numbers $\mathbb{N}$, the set of integers $\mathbb{Z}$, etc and an example of an uncountable set is the set of real numbers $\mathbb{R}$.

Countable Example

Theorem 1. *The set of integers $\mathbb{Z}$ is countable.*

Proof. To prove that $\mathbb{Z}$ is countable, we need to show that there exists a function $f(x) : \mathbb{Z} \rightarrow \mathbb{N}$ that is a bijection. We constructed the following function $f : \mathbb{Z} \rightarrow \mathbb{N}$ in class.

$$f(x) = \begin{cases} 2x & \text{if } x > 0 \\ -2x + 1 & \text{if } x \leq 0 \end{cases}$$

We convinced ourselves that this function is bijective in class using a combination of “laser vision”, “pattern matching”, and “proof by picture”. While that is great for intuition, we need to also learn how to prove it formally using math. We do that now.

To prove that $f(x)$ is a bijection, we need to show that $f(x)$ is one-one and onto.

One-one: To prove that $f(x)$ is one-one, we need to show that if $f(x_1) = f(x_2)$ then $x_1 = x_2$. We have several cases depending on whether x_1 and x_2 are positive or not.

Case 1. $x_1 > 0$ and $x_2 > 0$ then if

$$\begin{aligned} f(x_1) &= f(x_2) \\ 2x_1 &= 2x_2 \\ x_1 &= x_2 \end{aligned}$$

Case 2. $x_1 \leq 0$ and $x_2 \leq 0$ then if

$$\begin{aligned} f(x_1) &= f(x_2) \\ -2x_1 + 1 &= -2x_2 + 1 \\ x_1 &= x_2 \end{aligned}$$

Case 3. $x_1 > 0$ and $x_2 \leq 0$ then if

$$\begin{aligned} f(x_1) &= f(x_2) \\ 2x_1 &= -2x_2 + 1 \\ x_1 + x_2 &= 1/2 \end{aligned}$$

This case can never happen because x_1 and x_2 are integers and their sum cannot be a fraction. Thus in this case, the “if” part is false, so we do not need to prove the “then” part. The same happens when we try out the fourth case where $x_1 \leq 0$ and $x_2 > 0$, we find out that we can never have a case that $f(x_1) = f(x_2)$ for any x_1 and x_2 in this case.

Onto: To prove that $f(x)$ is onto, we have to prove that for every $y \in \mathbb{N}$, there is an $x \in \mathbb{Z}$ such that $f(x) = y$. We pick an arbitrary natural number y and divide this into two cases.

Case 1. y is an even natural number. Then $y = 2k$ for some natural number k (this is the definition of an even number). Then $x = k$ is a positive integer that maps to y , that is, $f(x) = 2k = y$.

Case 2. y is an odd natural number. Then $y = 2k + 1$ for some natural number k (this is the definition of an odd number). Then $x = -k$ is a negative integer that maps to y , that is, $f(x) = (-2)(-k) + 1 = 2k + 1 = y$.

Since $f(x)$ is one-one and onto, it is a bijection. □

0.1 Uncountable Example: Cantor’s Diagonalization

Theorem 2. *The power set $\mathcal{P}(\mathbb{N})$ of natural numbers $\mathbb{N}$ is not countable.*

Proof. When you have to prove a negative statement, such as this or in general prove that a statement X is false, it may be helpful to think about proof by contradiction first. In proof by contradiction, you start by assuming that the statement X is true and find an absurdity or a “contradiction”, which is a conclusion that violates one of your primary assumptions.

Here, let us assume that the power set $\mathcal{P}(\mathbb{N})$ is countable. If $\mathcal{P}(\mathbb{N})$ is countable, then by definition, there must exist a function $f : \mathcal{P}(\mathbb{N}) \rightarrow \mathbb{N}$ that is a bijection. This means that each subset $S \in \mathcal{P}(\mathbb{N})$ can be mapped to a natural number $j \in \mathbb{N}$, that is, $f(S) = j$ or in other words, this means that the subsets of $\mathbb{N}$ can be enumerated as a list $(S_{f(S)})_{S \in \mathcal{P}(\mathbb{N})} = S_1, S_2, S_3, \dots$, so that every subset is S_j for some $j \in \mathbb{N}$.

Consider the set $D = \{i \in \mathbb{N} \mid i \in S_i\}$. This set can be pictorially represented as the diagonal (blue cells) in the following table where a “Y” in a cell (S_i, j) means that element j is included in set S_i and a “N” in that cell means that element j is not included in set S_i .

$S_i \backslash j$	1	2	3	4	5	...
S_1	Y	N	N	Y	N	...
S_2	N	N	Y	N	N	...
S_3	Y	N	Y	N	Y	...
S_4	N	N	N	N	N	...
S_5	N	N	Y	Y	Y	...
...	...	...	...	...	...	...
S_k	N	Y	N	N	N	...

Let set $\bar{D} = \mathbb{N} - D = \{i \in \mathbb{N} \mid i \notin S_i\}$. This set is essentially formed by flipping the Ys and Ns in the diagonal. This set $\bar{D}$ is a subset of $\mathbb{N}$ so it must be equal to a set $S_k = \bar{D}$ for some natural number k . But this leads to the following contradictory statements.

1. If $k \notin \bar{D}$, then $k \in D$ and thus $k \in S_k = \bar{D}$.
2. If $k \in \bar{D}$, then $k \notin D$, thus $k \notin S_k = \bar{D}$.

Thus, we have our contradiction as k can either be in set $\bar{D}$ or not in set $\bar{D}$. Thus, our assumption that $\mathcal{P}(\mathbb{N})$ is countable is false, that is, $\mathcal{P}(\mathbb{N})$ is not countable.

□